This assignment will not be graded and is only for practice.

Partial Derivatives:

Level 1:

1) What is the rate of change of $f(x, y, z) = 5x^2 - 6xy + 3z^2 - 2yz$ at (1,2,-1) with respect to z?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -10

1 point

2) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x, y) = x^3 + 3xy$. Identify the correct option with respect to the partial derivatives of f. ***1 point***

- [ ] $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$
- [ ] $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0)$
- [ ] $\frac{\partial f}{\partial x}(1, 1) = \frac{\partial f}{\partial y}(1, 1)$
- [ ] $\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0)$
- [ ] $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} + 3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0)$
$\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0)$

3) Let $f(x, y) = \frac{xy}{x^2+y}$. Identify the correct options with respect to the partial derivatives of f. ***1 point***

- [ ] $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not defined at $(0, 0)$
- [ ] $\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 1/4$
- [ ] $\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 0$

☐ $\frac{\partial f}{\partial x}(1,0) = 0, \frac{\partial f}{\partial y}(1,0) = 1$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not defined at $(0,0)$
$\frac{\partial f}{\partial x}(1,1) = 0, \frac{\partial f}{\partial y}(1,1) = 1/4$
$\frac{\partial f}{\partial x}(1,0) = 0, \frac{\partial f}{\partial y}(1,0) = 1$

4) Let $f : \mathrm{R}^2 \to \mathrm{R}$ be defined as $f(x,y) = x^2 + 4xy - y^2$. Suppose $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $(12,k)$. Find the value of k

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Level 2:

5) Suppose $f(x, y, z) = xy + yz + zx - 3xyz$ is a multivariable function defined on domain $D \subset \mathbb{R}^3$. Which of the following is(are) correct? ***1 point***

- [ ] f is a scalar valued multivariable function.
- [ ] The rate of change of the function f at (0,1,1) with respect to x is 1.
- [ ] The rate of change of the function f at (2,2,2) with respect to x is same as the rate of change of the function f at (2,2,2) with respect to z.
- [ ] The rate of change of the function f at (1,3,5) with respect to y is -9.

No, the answer is incorrect.
Score: 0

Accepted Answers:

f is a scalar valued multivariable function.
The rate of change of the function f at (2,2,2) with respect to x is same as the rate of change of the function f at (2,2,2) with respect to z.
The rate of change of the function f at (1,3,5) with respect to y is -9.

6) Suppose $f : D \longrightarrow \mathbb{R}$ is a multivariable function defined on domain $D \subset \mathbb{R}^2$ and also satisfying the following conditions: ***1 point***

(i) $\frac{\partial f}{\partial x} = 6x + 4xy + 3y$
(ii) $\frac{\partial f}{\partial y} = 2x^2 + 3x - 3y$
Which of the following functions can not be f?

- [ ] $f(x, y) = 3x^2 + 2x^2y + 3xy - \frac{3}{2}y^2$
- [ ] $f(x, y) = 3x^2 + 2x^2y - 3xy - \frac{3}{2}y^2$
- [ ] $f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2$
- [ ] $f(x, y) = 3x^2 + 3y^2 + 3xy$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$f(x,y) = 3x^2 + 2x^2y - 3xy - \frac{3}{2}y^2$
$f(x,y) = 3x^2 - 3xy - 2x^2y + 3y^2$
$f(x,y) = 3x^2 + 3y^2 + 3xy$

7) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(x,y) = (P(x,y), Q(x,y))$, where $P(x,y) = 3x^2y$, $Q(x,y) = 5x + y^3$. Consider a matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$. If the determinant of A is $axy^3 - bx^2$ where $a, b \in \mathbb{R}$, then find the value of $a - b$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

8) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(x,y) = (P(x,y), Q(x,y))$, where $P(x,y) = 6x + y^2$, $Q(x,y) = 3x^2 + 2y$. Consider a matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$. If $det(A) = 0$, then find the value of xy

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

9) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x,y) = e^{4(x+y)} sin(7x - 4y)$. If $4\frac{\partial f}{\partial x} + 7\frac{\partial f}{\partial y} = k.f(x,y)$ where $k \in \mathbb{R}$, then find the value of k.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 44

1 point

Your score is: 0/9